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REVIEW ARTICLE

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Neurotoxicity associated with acyclovir and valacyclovir: A systematic review of cases

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Abstract

What is known and objective: Acyclovir and valacyclovir are commonly used antivirals with good general tolerance. Despite their good safety profile, they can cause systemic adverse effects, such as neurotoxicity, which are less frequent and known. The objective of this review was to collect all the reported cases of neurotoxicity associated with acyclovir and valaciclovir published in the literature and characterize their clinical course and interventions.

Methods: A systematic review of cases was carried out following the guidelines established by "Preferred Reporting Items for Systematic Reviews and Meta-Analyses" (PRISMA). The research was carried out using the PubMed-Medline and Embase databases, between July 1984 and March 2021.

Results and discussion: A total of 119 cases with neurotoxicity mainly related to acyclovir (*n* = 88; 73.9%), followed by valaciclovir (*n* = 35; 29.4%) were analysed. 49.6% (*n* = 59) were men with a mean age of 59.5 years ± 21.1 (0.5–88). In 83.3% of the cases, renal impairment was documented and 57.1% (*n* = 68) with end-stage renal disease. The administered dose was higher than the renal adjustment recommendations in 59.7% of the cases. The global mean of onset of symptoms was 3.1 days ± 4.3 (0.2–28) after the start of antivirals. The mean recovery time was 9.8 days ± 21.7 (0.2–180). 74.4% of the patients had a recovery of ≤7 days, 15.9% between 8 and 15 days and 9.8% > 15 days.

What is new and conclusion: The neurotoxicity induced by acyclovir and its derivative valacyclovir is a poorly known and rare adverse effect that can occur mainly in patients with advanced age and impaired renal function. The most characteristic symptoms are confusion, altered level of consciousness, hallucinations, agitation and dysarthria. The basis of treatment is the discontinuation of the antiviral, and in some cases, it may require additional clearance by dialysis.

KEYWORDS
acyclovir, dialysis, neurotoxicity, renal failure, valacyclovir

1 | WHAT IS KNOWN AND OBJECTIVE

The varicella-zoster virus (VZV) included within the herpesviridae family is the cause of a high prevalence of infections in humans.[1] Acyclovir is a guanosine nucleoside analog approved in the 1970s (1974), widely used in the treatment of VZV and herpes simplex virus (HSV). It also has inhibitory capacity in vitro in the replication of other viruses such as Epstein-Barr virus (EBV) and cytomegalovirus (CMV). To develop its activity, it needs to be converted into its active form, acyclovir triphosphate, which is incorporated into the DNA chain and inhibiting DNA polymerase.[2] There are presentations for intravenous and oral administration. The latter has a considerably low oral bioavailability, around 15%–30% of the administered dose. The volume of distribution of acyclovir molecule is large allowing good tissue penetration, including the central nervous system (CNS), reaching high levels in the cerebrospinal fluid (CSF). Its elimination is mainly renal through filtration and tubular excretion; therefore, kidney damage can cause drug accumulation and systemic adverse effects. 9-carboxymethyl methylguanine (CMMG) is the only metabolite of acyclovir with 10%–15% renal elimination.[3]

In 1995, valacyclovir was approved by the FDA. It is the L-valine ester of acyclovir, which after oral administration suffers a first-pass effect on the liver and is rapidly and extensively converted to acyclovir. The main advantage of valacyclovir over Acyclovir is its oral bioavailability, which allows it to achieve plasma levels of acyclovir 3 to 5 times higher. In consequence, it is possible to use less frequent dose regimens but retaining the highly acceptable safety profile established for acyclovir.[4,5]

Both are drugs frequently used and with good general tolerance. Despite their good safety profile, they can cause systemic adverse effects. The most frequent toxicity is acute renal failure associated with the deposit of crystals in the renal tubule.[6] Another lesser known but no less important side effect is neurotoxicity, characterized by confusion, dizziness, drowsiness, stupor, coma and psychiatric symptoms, among others. The reported incidence is very low, and the main associated risk factors are renal deterioration, high doses of antivirals and age.[7] On the other hand, the presence of CMMG in cerebrospinal fluid has been related to the appearance of psychiatric symptoms.[8]

The objective of this review was to collect all the reported cases of acyclovir- and valacyclovir-associated neurotoxicity published in the literature and to analyse the demographic, clinical and neurological characteristics collected.

2 | METHODS

2.1 | Study selection

A systematic review of cases was carried out following the guidelines established by "Preferred Reporting Items for Systematic Reviews and Meta-Analyses" (PRISMA). A retrospective observational study was followed with all the data obtained. The research was carried out using the PubMed-Medline and Embase databases between July 1984 and March 2021, using the following search criteria: (acyclovir and neurotoxicity) OR (valacyclovir and neurotoxicity) OR (acyclovir and neuropsychiatric symptoms) OR (valacyclovir and neuropsychiatric symptoms) OR (acyclovir and epilepsy) OR (valacyclovir and epilepsy) OR (acyclovir and altered mental status) OR (valacyclovir and altered mental status) OR (acyclovir and encephalopathy) OR (valacyclovir and encephalopathy). Two authors (DBN and MCS) independently and in parallel conducted the initial title/abstract selection from original articles. Articles with no limits on language were reviewed. For studies not located or with incomplete information, corresponding authors were contacted. We included all published studies (clinical cases and case series) with cases of neurotoxicity related to the administration of acyclovir or valacyclovir and that provided data associated with neurological adverse effects. Articles in which neurotoxicity could be caused by other drugs, animal studies, brief communications to congresses, reviews or articles with incomplete data were excluded. Risk of bias was assessed using the Cochrane Collaboration tool.

2.2 | Data extraction and variables

DBN and MCS parallel and independently were responsible for extracting data from publications. Variables collected and analysed were age, sex, renal dysfunction, creatinine, urea, creatinine clearance (CrCl), concomitant nephrotoxic drugs, kidney transplant, correct dose adjusted to CrCl, obesity, antiviral exposure time until symptom onset, grade of neurological recovery, time to full recovery of symptoms, stay in the critical care unit, death, interventions: withdrawal of antiviral, dialysis, peritoneal dialysis or extracorporeal renal replacement therapies (ERRT). Also, the number of cases associated with acyclovir and/or valacyclovir, route of administration, indication, total daily dose, levels of acyclovir and determination of 9-carboxymethoxymethylguanine (CMMG) metabolite in serum and CSF. In addition, neurological clinical characteristics such as encephalopathy, epileptic status, seizures/myoclonus, disorientation, hallucinations, tremor, asterixis, ataxia, aphasia, dysarthria, agitation, photophobia, decreased level of consciousness, delirium, peripheral neuropathy, dizziness, depression, insomnia and irritability were collected, among others.

Renal dysfunction was defined as the presence of at least 1 of the following criteria: glomerular filtration rate less than 60 ml/min, serum creatinine greater than 2 mg/dl, or a description in the text such as kidney failure or chronic kidney failure. End-stage renal disease (ESRD) was defined as the presence of renal replacement therapy or a text description. The "correct dose adjusted to renal function" was considered to be the dose received in accordance with the recommendations set out in Table 1. A "decreased level of consciousness" was defined as a reduction in alertness, which includes sleepiness, drowsiness and stupor. The "coma" state was recorded as another independent variable. The variable seizures / myoclonus was considered as any involuntary movement or fasciculation, presence of alteration in electroencephalogram

TABLE 1 Recommended acyclovir and valacyclovir dosage based on renal function

	Acyclovir		Valacyclovir
	Intravenous	Oral	
CrCl >50 ml/min	5-15 mg/kg/8 h	200-800 mg/5 times a day	1 g/8 h
CrCl 25–50 ml/min	5-10 mg/kg/12 h	200-800 mg/5 times a day	1 g/12 h
CrCl 10–25 ml/min	5-10 mg/kg/24 h	800 mg/8 h	1 g/24 h
CrCl <10 ml/min	2.5-5 mg/kg/24 h	800 mg/12 h	500 mg/24 h
Dialysis	5 mg/kg/24 h	400-800 mg/24 h	500 mg/24 h, after dialysis session.
ERRT	7.5-10 mg/kg/24 h	600-1200 mg/24 h	500 mg/24 h
Peritoneal dialysis	5 mg/kg/24 h	400-800 mg/24 h	500 mg/24 h

Abbreviations: CrCl, creatinine clearance; ERRT, extracorporeal renal replacement therapies.

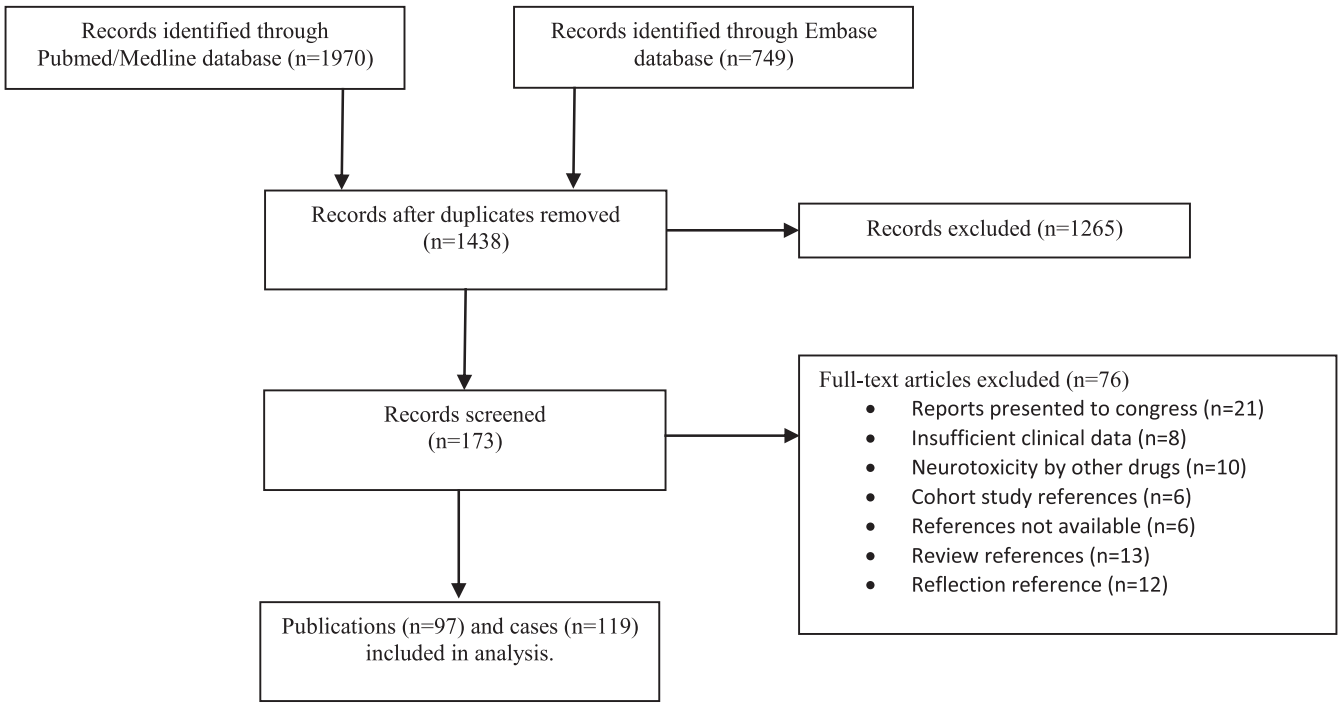


FIGURE 1 PRISMA flow diagram of the study selection

(EEG) or through text documentation. The peripheral neuropathy variable was identified as the presence of neuropathic pain, sensory or muscular alterations or through text documentation.

2.3 | Statistical analysis

A descriptive analysis was performed using the SPSS program (version 20.0), and the qualitative variables were expressed in percentages and the quantitative variables in mean, standard deviation and range.

3 | RESULTS

During this research, a total of 2719 articles were identified. Ninety-seven of those were included in the analysis, representing a total of 119 independent patients (Figure 1). The cases were extracted from 86 clinical cases[9-35,37-48,50-54,56-63,65-71,73,75-77,80-83,85-100,102,103] and 11 case series.[36,49,55,64,72,74,78,79,84,101,113] Figure 1 shows the case selection flow diagram.

The demographic and clinical characteristics of the 119 patients with acyclovir- and valacyclovir-associated neurotoxicity can be seen in Table 2. In most cases, neurotoxicity was related to acyclovir (*n* = 88; 73.9%), followed by valacyclovir 35 (29.4%). The oral route was the most frequent form of administration (*n* = 85; 71.4%). The population mean age was 59.5 years ± 21.1 (0.5–88), with 9.2% (*n* = 11) of paediatric patients (<18 years)[38,53,69,70,81,84-86,93] and 61.3% (*n* = 73) of patients older than 65 years.

In 83.2% of the cases, renal impairment was documented at the time of administration, and in 57.1% (*n* = 68), they presented ESRD or undergoing renal replacement therapy. Serum creatinine was recorded in 47 patients, with a mean of 5.6 mg/dl ± 3.6 (0.6–17) and

TABLE 2 Demographic and clinical characteristics of acyclovir and valacyclovir neurotoxicity patients

	N = 119
Acyclovir, n (%)	88 (73.9)
IV, n (%)	57 (47.9)
Oral, n (%)	
Valacyclovir, n (%)	35 (29.4)
Acyclovir and valacyclovir, n (%)	12 (10.1)
Mean age, years (± SD)	59.5 ± 21.1
Male gender, n (%)	59 (49.6)
Renal dysfunction, n (%)	99 (83.2)
Dialysis	53 (44.5)
ERRT	3 (2.5)
Peritoneal dialysis	15 (12.6)
Kidney transplant, n (%)	11 (9.2)
Mean cr, mg/dl (± SD)	5.6 ± 3.6
Mean crcl, ml/min (± SD)	34.5 ± 34.7
Mean urea, mg/dl (± SD)	89.0 ± 53.5
Dose not adjusted to crcl, n (%)	71 (59.7)
Co-administered nephrotoxic drugs, n (%)	20 (16.8)
Co-administered nephrotoxic drugs, n (%)	23 (19.3)
Obesity, n (%)	5 (4.2)
Mean of neurotoxicity onset time, days (± SD)	3.1 ± 4.3
Treatment, n (%)	
Antiviral withdrawal	98 (82.3)
Received dialysis	61 (51.3)
Received ERRT	4 (3.4)
Received peritoneal dialysis	11 (9.2)
Complete recovery, n (%)	111 (93.3)
Critical care stage, n (%)	26 (21.8)
Death, n (%)	5 (4.2)
Mean time to clinical improvement, days (± SD)	
All patients	9.8 ± 21.7
Treatment whit dialysis	5.9 ± 6.8
Mean of total daily iv acyclovir dose, mg (± SD)	2121.2 ± 1553.8
Mean of total daily oral acyclovir dose, mg (± SD)	2251.9 ± 1411.4
Mean of total daily valacyclovir dose, mg (± SD)	2488.9 ± 1291.2
Mean of acyclovir serum levels, mcg/ml (± SD)	
Indication, n (%)	
HSV	26 (21.8)
VZV	86 (72.3)
HSV prophylaxis	2 (1.7)
CMV	4 (3.4)
EBV	1 (0.8)

Abbreviations: CMV, cytomegalovirus; Cr, creatinine; CrCl, creatinine clearance; EBV, Epstein-Barr virus; ERRT, extracorporeal renal replacement therapies; HSV, herpes simplex virus; IV, intravenous; VZV, varicella-zoster virus.

CrCl in 23, with a mean of 34.5 ml/min ±34.7 (5–131). On the other hand, the administered dose was incorrect in more than half of the patients (59.7%), being higher than the recommendations for renal adjustment. In 2 obese patients,[25,94] the dose of intravenous acyclovir was estimated according to the real weight instead of the adjusted ideal weight with the consequent overdose.

The global mean onset of symptoms was 3.1 days ± 4.3 (0.2–28) after the start of antivirals. All patients suffered neurological alterations, and the majority recovered their baseline state (93.3%). Only 3 cases had an incomplete reversal of symptoms.[61,75,93] Lastly, 5 patients died, of those, 4 were older than 65 years with significant comorbidity and 3 presented ESRD under renal replacement therapy. The last dead patient was only 29 years old however diagnosed with leukaemia and under treatment with chemotherapy.[12,18,69,78,84]

The neurological symptoms collected are described in Figure 2. The most frequently described symptoms were disorientation (*n* = 69; 58%), decrease level of consciousness (*n* = 45; 37.8%), hallucinations (*n* = 46; 36.1%), agitation, (*n* = 32; 26.9%) and dysarthria (*n* = 23; 19.3%). In 5 cases,[21,27,48,54,89] although the authors attributed the neurological symptoms to antivirals, viral encephalitis cannot be definitively ruled out, since a positive result was obtained in the PCR of HZV in CSF (Table 3).

Interventions in the treatment of neurotoxicity included antiviral discontinuation (*n* = 98; 82.3%), dose adjustment to renal function (*n* = 6; 5.0%), clearance of acyclovir by dialysis (*n* = 61; 51.3%), peritoneal dialysis (*n* = 11; 9.2%) and ERRT (*n* = 3; 2.5%). The mean recovery time was documented in 100 cases, with a mean of 9.8 days ± 21.7 (0.15–180). 74.4% of the patients had a recovery of ≤7 days, 15.9% between 8 and 15 days and 9.8%> 15 days.

Detection of acyclovir levels in plasma was requested in 53 cases (44.5%), with a mean of 17.2 mcg/ml ±25.3, and levels in CSF in 10 cases,[15,17,18,26,31,113] with a mean of 1.2 mcg/ml ±1.6. The CMMG metabolite was also obtained in 10 cases,[25,31,34,37,51,113] with 8 determinations in plasma, 8 in CSF and 1 in urine.

4 | DISCUSSION

This work is the first systematic review, to our knowledge, that joins up and describes the neurotoxicity associated with acyclovir and its derivative, valacyclovir published in the literature. It was found that age (> 65 years) and renal impairment, especially ESRD, behaved as the main related risk factors. As already commented, the clearance of acyclovir is fundamentally renal and crosses the blood-brain barrier; therefore, a decrease in glomerular filtration can increase CSF levels and produce neurotoxicity.[4] Despite the fact that the neurotoxicity mechanism is not completely clear, a proposed hypothesis would be the cellular alteration through the inhibition of DNA polymerase at the mitochondrial level.[104]

An alternative treatment in patients with mucocutaneous HSV or VZV and risk factors associated with neurotoxicity could be famciclovir. It has been used previously with success in patients who developed renal toxicity secondary to acyclovir, and although

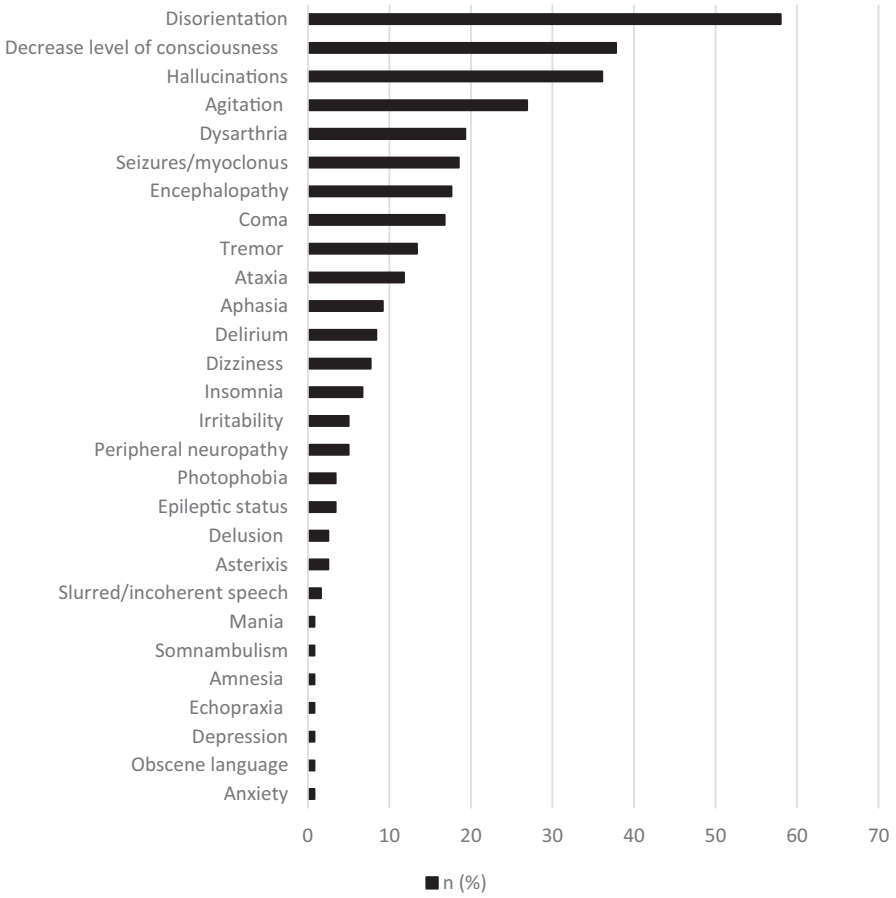


FIGURE 2 Clinical manifestations of acyclovir and valacyclovir neurotoxicity were tabulated and reported as a percentage of the total cohort (*n* = 119), from most to least frequently observed

TABLE 3 Acyclovir- and valacyclovir-induced neurotoxicity: A clinical picture

Risk factors	Signs and symptoms	Interventions
• Renal dysfunction • Older age • Drug overdose	• Disorientation • Decrease level of consciousness • Hallucinations • Agitation • Dysarthria • Coma • Seizures/myoclonus • Encephalopathy • Tremor • Ataxia • Delirium • Aphasia • Dizziness • Irritability • Insomnia • Peripheral neuropathy	• Drug discontinuation • Dialysis • Peritoneal dialysis • ERRT

Abbreviation: ERRT, extracorporeal renal replacement therapies.

it is not without neurological adverse effects, its incidence seems more limited, with only one case of neurotoxicity reported to date.[105,106]

It came as a surprise to observe that the administered dose was higher than the renal adjustment recommendations in more than half of the patients. This can be partly explained by the fact that renal function is dynamic and in some patients the test was requested when the neurotoxicity was already established, and errors in the calculation of the initial dosage. On the contrary, the other half of the patients developed neurotoxicity despite an apparently adequate adjustment. Watson Wa, et al, in a case of this series, used a Bayesian method to adjust the dose and showed an overestimation of the real renal function when estimated with a traditional population formula.[21] Therefore, caution can be advised in the use of traditional formulas to evaluate renal function due to their limitations in elderly patients or those with renal failure.[107]

In patients with HZV treated with acyclovir or valacyclovir and neurological symptoms, it is very difficult to establish a correct differential diagnosis of neurotoxicity with respect to viral encephalitis. In this study, it has been observed that the most frequent neurological manifestations were psychiatric symptoms, confusion, altered level of consciousness and hallucinations and dysarthria. This symptomatology is compatible with other reviewed studies with patients with neurological toxicity secondary to both antivirals.[36,72,108] However, viral encephalitis is characterized by the presence of fever, headache, meningeal symptoms, such as a stiff neck and cranial neuropathy. Furthermore, the presence of pleocytosis and a positive PCR or antibodies to HZV in CSF are indicators of viral infection.[109] The onset of neurotoxicity symptoms in our population was 3 days after starting the antiviral, whereas encephalitis appeared 7 days after the skin lesions appeared.[75] On the opposite, although in most patients the toxicity was reversible with a recovery equal to or less than 7 days, it was reported considerable severity, with 26% of the cases admitted to the critical care unit and 4.2% mortality.

Withdrawal or discontinuation of the antiviral is the basis of treatment, as it has been described in the vast majority of our patients. On the other hand, acyclovir is almost 50% dialysed due to its low molecular weight, low protein binding and high solubility.[110] HD was the treatment used in almost half of the patients, and a clearance of 40%–60% of the antiviral level has been documented with sessions of 3-5 h of HD.[24,26,35,110,111] Another treatment technique used was PD. Despite the fact that HD appears more effective than PD in extraction, good results were obtained in reversing neurological symptoms with intensified PD.[33] Finally, another extraction method described with therapeutic utility was ERRT.[12,17,20]

Regarding acyclovir levels, in almost half of the cases it was documented that they were obtained. Its relationship with neurotoxicity is yet to be clarified, since there are cases of neurotoxicity that are supposedly in range. The delay in the extraction of the sample has been sought as an explanation. Despite that, acyclovir levels do not appear to be a test of usefulness at present.[13,14,27,52,76]

Acyclovir is metabolized by the enzymes alcohol dehydrogenase and aldehyde dehydrogenase, to its main metabolite, 9-carboxymethoxymethylguanine (CMMG). Helldén et al, demonstrated the correlation between the accumulation of the CMMG metabolite and the development of neuropsychiatric symptoms in patients with CKD treated with acyclovir or valacyclovir.[8,112,113] In consequence, the determination of CMMG can be useful in the diagnosis of neurotoxicity.

5 | LIMITATIONS

This study is susceptible to several limitations that deserve comment. In the first place, since it is a review made from clinical cases and case series, it is subject to notification and publication bias. The data obtained come from subjects who presented toxicity induced by very similar, but not identical drugs, with different dosages and routes of administration. The evaluation of renal function and the calculation of the dose adjustment were performed inconsistently, which limits the confirmation of renal failure and overdose, as risk factors related to neurotoxicity. Finally, the observational design of our study describes the documented interventions in the management of neurotoxicity but does not allow us to conclusively evaluate their efficacy. Despite its limitations, this study provides a complete compilation that summarizes and describes the characteristics of the neurotoxicity associated with acyclovir and its derivative.

6 | WHAT IS NEW AND CONCLUSION

The neurotoxicity induced by acyclovir and its derivative valacyclovir is a poorly known and rare adverse effect that can occur mainly in patients with advanced age and impaired renal function. The most characteristic symptoms are confusion, altered level of consciousness, hallucinations, agitation and dysarthria. Toxicity appears after 3 days of treatment and reverses completely after 7 days or less. The basis of treatment is the discontinuation of the antiviral and in some cases an additional intervention such as dialysis may be needed. To minimize the risk of developing neurotoxicity in patients with risk factors, we recommend monitoring renal function to make a correct dose adjustment and avoid overdoses. Determination of the metabolite CMMG can be useful in the diagnosis of neurotoxicity. To conclude, more studies are needed to clarify the role of monitoring acyclovir plasma levels in the diagnosis of neurological toxicity.

CONFLICT OF INTEREST

None declared.

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